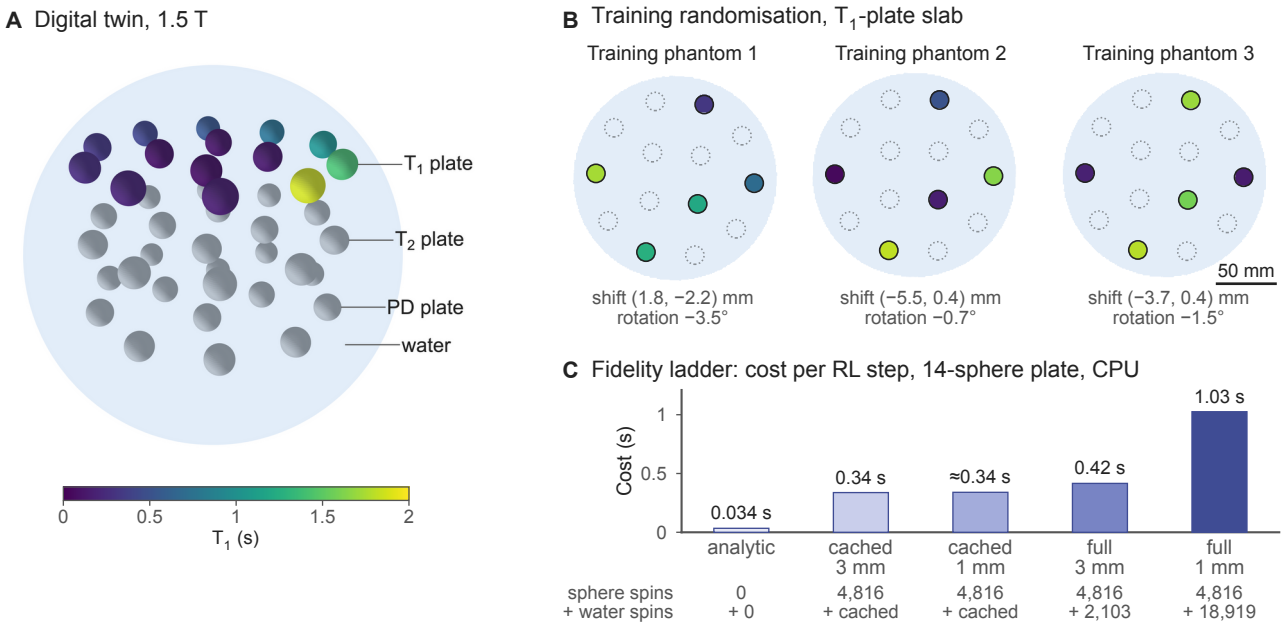

A Digital twin, 1.5 T
T1 plate
T2 plate
PD plate
water
0
0.5
1
1.5
2
T1 (s)
B Training randomisation, T1-plate slab
Training phantom 1
Training phantom 2
Training phantom 3
50 mm
shift (1.8, −2.2) mm
rotation −3.5°
shift (−5.5, 0.4) mm
rotation −0.7°
shift (−3.7, 0.4) mm
rotation −1.5°
C Fidelity ladder: cost per RL step, 14-sphere plate, CPU
Cost (s)
1
0.5
0
0.034 s
0.34 s
≈0.34 s
0.42 s
1.03 s
analytic
cached 3 mm
cached 1 mm
full 3 mm
full 1 mm
sphere spins
+ water spins
0
+ 0
4,816
+ cached
4,816
+ cached
4,816
+ 2,103
4,816
+ 18,919